



Instruments Designed for Teaching

The Green-Laser Upgrade to **FARADAY ROTATION**

A PRODUCT OF TEACHSPIN, INC.

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0. Introduction

This slim Manual describes the ‘green-laser upgrade’ to TeachSpin’s ‘Faraday Rotation’ apparatus. We assume you have that apparatus, and have the standard red-laser light source that comes with it. We also assume that you’ve understood the theoretical motivation for performing Faraday-Rotation experiments, and that you’ve performed at least the basic demonstration experiments with red light. This Manual, then, takes up the motivation for trying a different *color* of light, and discusses how to do that using the green-laser upgrade.

Section 1 takes up the ‘Verdet constant’ for Faraday Rotation, and the simplest theory of how it should vary with the wavelength λ of the light used. Section 2 takes up the practical issues of how you find refractive-index information $n(\lambda)$ for a given sample, and how you can use it to get the quantity $\lambda \cdot dn/d\lambda$ you will want.

Section 3 describes how to use the green-laser source in place of the red laser, and Section 4 describes a complication – and its cure – arising when electronic (rather than direct visual) detection is used. This is where the ‘low-pass filter’ item of the upgrade is discussed.

Section 5 makes explicit a sequence of methods-of-operation for the Faraday-Rotation experiment that are left implicit in the main Faraday-Rotation Manual. In particular, this section provides a sort of bridge that can take students from the most straightforwardly geometrical, and direct, demonstrations and measurements of Faraday rotation, to more abstract but also more sensitive and precise methods for detecting Faraday rotations, especially when such rotations are very small.

Appendix A1 is a tutorial ‘case study’ on how the simplest low-pass filter works, and how one goes about designing such an item. Appendix A2 discusses the actual wavelengths of the light emerging from the red and green laser modules. Finally, Appendix 3 derives the ‘effective sample length’ for the use of solid, and liquid-cell, samples in the solenoid in the apparatus.

The Manual includes a few boxed Exercises that students ought to undertake in their reading, for orientation to orders-of-magnitude, units, and the like.

1. Theory of the Verdet Constant

If you're familiar with Faraday Rotation, then you know the 'observables'. In the simplest experiments, you shine linearly-polarized light of wavelength λ through a length L of transparent sample. You immerse that sample in a magnetic field B , whose direction is along the light beam. Finally, you find that the output light is also linearly polarized – but at a polarization angle that's been *rotated* through an angle θ . You should not be surprised to hear that θ scales with L , and you can check that it scales with B as well. What's left is called the Verdet constant V of the sample material, defined via

$$\theta = V B L \quad .$$

The units of V are radians (of rotation) per tesla (of field) $\times$ meter (of sample), or rad/(T·m). (Older references will use instead the units of arc-minutes for rotation, per gauss·cm of field $\times$ length.)

This model defines V , and it 'scales out' the B - and L -dependence, but it does not connect V to anything more basic. Neither does it say whether, or how, V might depend on the wavelength of the light used. A deeper theory for V depends on two theoretical constructions.

First, as used by Fresnel in the (earlier) theory of optical rotation by chiral samples:

One resolves the input light's linear polarization into a superposition of equal amplitudes of the two circular-polarization components. Then one assigns two distinct indices of refraction, n_+ and n_- , to the two circular polarizations. [These are expected to differ if the sample is 'chiral', i.e. if its molecules have 'handedness' and respond differently to right- and to left-handed circularly-polarized light.] Each circularly-polarized component propagates independently through the sample, each picking up a net phase according to its own index of refraction. At the exit of the sample, one still has two equal-amplitude circular-polarization components, but they've accumulated *different* phase shifts. So the sum of these two components, at the end of the sample, is again a linear polarization – but one that's been rotated by angle

$$\theta = \pi \frac{L}{\lambda} (n_+ - n_-) \quad .$$

Notice this argument does *not* predict how different n_+ and n_- might be, nor even why they might be different. But it does say that in giving a prediction for θ , the difference $(n_+ - n_-)$ gets multiplied by π (radians) and also by the quotient L/λ . For typical experiments, $L \approx 10 \text{ cm} = 10^{-1} \text{ m}$, while for visible light, $\lambda < 1 \text{ } \mu\text{m} = 10^{-6} \text{ m}$, so this factor is $> 10^5$. Hence even a *tiny* 'circular birefringence' $n_+ - n_-$ can lead to readily-detectable effects.

Second, as contributed by Becquerel, a model suggesting why n_+ and n_- ought to differ, even for a non-chiral sample, in the presence of a magnetic field:

This calculation depends on a model for the interaction of light and matter, and Becquerel's model involved the then-new 'electron theory' of atoms of matter. In that era models were classical, though now we use quantum-mechanical models instead. But the conclusions are similar, and are related to the Zeeman effect: because the presence of B can and does shift atomic energy levels, the absorption and emission spectrum of a sample changes with B . But the dispersion of a sample is connected to its absorption; hence there's an effect of B on index of refraction n . Becquerel's model showed how the indices of refraction for the two circular-polarization components would be affected differently, and it gave

$$n_+ - n_- = -\frac{eB}{2\pi m_e c} \cdot \lambda^2 \frac{dn}{d\lambda} .$$

Here e is the fundamental charge, and m_e gives the electron mass.

Combining this result with Fresnel's, we get a prediction for Faraday rotation of

$$\theta = \pi \frac{L}{\lambda} \cdot \frac{eB}{2\pi m_e c} \cdot \lambda^2 \frac{dn}{d\lambda} = V B L , \quad \text{with} \quad V = \frac{e}{2m_e c} \cdot \lambda \frac{dn}{d\lambda} .$$

We have *not* kept track of signs here. Notice that the B - and L -dependencies have both emerged naturally, so that we can identify what's left as the Verdet constant V ; and note that V is predicted to depend on the (universal) constant $e/(2m_e c)$ and on the (sample-dependent) function $\lambda \cdot dn/d\lambda$.

Exercise: Evaluate $e/(2m_e c)$, and show that it can be written as $X \text{ rad}/(\text{T}\cdot\text{m})$. Find the number X . Also find the dimensionality of the factor $\lambda \cdot dn/d\lambda$.

Now Becquerel's theory, or the simplest quantum-mechanical model that is analogous to it, is too simple to describe the details of the effect of a magnetic field on the atoms and molecules in (say) optical glass. But to a surprising degree, all the differences between the simple models and the real thing can be captured in a single 'magneto-optical constant' γ , whose value will depend on the sample, but not (we hope) on the wavelength of light used. Then we'd have

$$\theta = V B L , \quad \text{with} \quad V = \gamma \cdot \frac{e}{2m_e c} \cdot \lambda \frac{dn}{d\lambda} .$$

In this model, if we have $n = n(\lambda)$ 'dispersion information' for a sample, we can compute the $\lambda \cdot dn/d\lambda$ function, and we can evaluate this quantity for any given wavelength. So if we measure V at that wavelength, we determine γ in the equation above. And once we have γ 's value, we have

$$V = V(\lambda) = \gamma \cdot \frac{e}{2m_e c} \cdot \lambda \frac{dn}{d\lambda}$$

as a fully-predictive formula, claiming to give the Verdet constant's value for any wavelength.

2. Theory of optical dispersion $n(\lambda)$

The theory of Section 1 for the Verdet constant depends on having a model for $n(\lambda)$ for a sample – this is the wavelength dependence of the index of refraction, or the ‘dispersion curve’, for the material. But Section 1 did not work out $n(\lambda)$ for any particular sample. This can actually be done, from first principles, for samples that are simple enough. The index of refraction of a material arises because the sample is polarizable; that is, its atoms develop electric dipole moments in proportion to the electric field of the light illuminating them. What’s more, the polarization of the atoms is frequency-dependent, growing as the frequency of the light used approaches a resonant frequency of the atoms in the sample.

For an atom as simple as He, for a sample of gas-phase helium, these calculations can be done from quantum-mechanical first principles. For most materials, there’s little incentive to work from fundamentals. But still we realize that index of refraction is connected to polarizability; that polarizability is expected to show resonant behavior; and that such resonances correspond to absorption of the light by the sample. Hence dispersion of a sample is characteristically connected to absorption.

Helium atoms can’t absorb low-frequency light, as their lowest excited states lie nearly 20 eV in energy above the ground state.

Exercise: What wavelength of light has photons of energy 20 eV? Where in the electromagnetic spectrum does this wavelength fall?

If helium has its lowest-energy absorption this high in energy, then in the visible region of the spectrum it shows negligible absorption. Yet it still exhibits dispersion, and its index of refraction is predictable, and predicted to be smaller at lower frequencies, and to grow as frequencies rise (toward those much higher frequencies where resonances, and absorption, occur). So the higher visible frequencies, i.e. the shorter wavelengths, of visible light ought to exhibit larger n -values.

What’s true for helium gas is true for almost any sample that’s transparent in the visible. Such a sample will surely absorb light of some higher frequency, perhaps one in the ‘near ultraviolet’ (with λ just beyond the violet end of the spectrum, say in the 300 – 400 nm range). Then we expect $n(\lambda)$ to be largest in the violet, and to be decreasing toward the red end of the spectrum. So we expect ‘normal’ dispersion, $n = n(\lambda)$, with n decreasing as λ increases.

The ‘resonance model’ even predicts some forms for $n(\lambda)$. An early empirical model due to Cauchy is

$$n(\lambda) = A + \frac{B}{\lambda^2} ,$$

which predicts n rising to infinity as $\lambda \rightarrow 0$. In reality, n grows large before λ goes all the way to zero; but even so, this model predicts

$$\frac{dn}{d\lambda} = 0 - \frac{2B}{\lambda^3} \quad , \quad \text{and so} \quad \lambda \frac{dn}{d\lambda} = -\frac{2B}{\lambda^2} \quad .$$

That, in turn, predicts a Verdet constant which is larger for smaller wavelengths: $V = V(\lambda)$.

Real materials of technological interest have their $n(\lambda)$ curves measured, not predicted. For pure crystalline NaCl, for example, this curve is as well-known, and it is just as quantum-mechanically controlled as is the mass density ρ ; but no one would compute either from first principles, given how easy they are to measure.

Similarly, any optical glass worthy of a standard manufacturer's recipe has a characteristic $n(\lambda)$ curve, and real optical glasses have their production so tightly controlled that $n(\lambda)$ is reproducible to precision of order 10^{-6} . The $n(\lambda)$ values are measured empirically at standard wavelengths (namely, those of atomic spectral lines) and are available in the form of tables.

That tabular information, in turn, can be represented by empirical models. A typical model is of Sellmeier form, one term of which has the form (in terms of optical frequency f)

$$n^2 - 1 = \frac{\alpha}{\beta^2 - f^2} \quad .$$

Note that this model predicts $n^2 - 1$ rather than n itself. The form of the model comes from resonance theory, and it's clear that this one-term model corresponds to a resonance at frequency $f = \beta$. The models are typically translated into dependence on wavelengths rather than frequencies, so the terms have the form

$$n^2 - 1 = \frac{\alpha}{\beta^2 - (c/\lambda)^2} = \frac{\alpha\lambda^2}{\beta^2\lambda^2 - c^2} = \frac{(\alpha/\beta^2)\lambda^2}{\lambda^2 - c^2/\beta^2} = \frac{B\lambda^2}{\lambda^2 - c} \quad .$$

Typical optical glasses are modelled by in a three-term Sellmeier expansion, assuming two resonances in the ultraviolet, and one in the infrared:

$$n^2(\lambda) - 1 = \frac{B_1\lambda^2}{\lambda^2 - c_1} + \frac{B_2\lambda^2}{\lambda^2 - c_2} + \frac{B_3\lambda^2}{\lambda^2 - c_3} \quad .$$

One advantage of such an analytical model is that (unlike tabulated information) it permits differentiation. We get, for example, in a one-term expansion

$$\frac{d}{d\lambda}(n^2 - 1) = 2n(\lambda) \frac{dn}{d\lambda} = \frac{d}{d\lambda} \left(\frac{B\lambda^2}{\lambda^2 - c} \right) = \frac{-2BC\lambda}{(\lambda^2 - c)^2} \quad ,$$

and this allows $dn/d\lambda$, and from it $\lambda \cdot dn/d\lambda$, to be calculated for any desired wavelength.

TeachSpin's 'Faraday Rotation' experiment is shipped with one standard glass sample, of length $L = 100 \pm 1$ mm, made from optical glass with the Schott number SF-57. This glass is chosen

because it has a rather large index n , rather high dispersion $dn/d\lambda$, and rather high Verdet constant. The tabulated data for SF-57's index of refraction is

wavelength λ (in μm)	index n
0.4358	1.89391
0.4800	1.87425
0.4861	1.87205
0.5461	1.85504
0.5893	1.84636
0.6328	1.83957
0.6438	1.83808
0.6563	1.83651
0.7065	1.83104
0.8521	1.82045
1.0600	1.81184

where $n = n(\lambda)$ values are given as pure numbers, and λ 's are given in units of μm . An empirical model for this tabulated data is given by a three-term Sellmeier expansion with coefficients

B_1	1.8165 1371
B_2	0.4288 9364
B_3	1.0718 6278
C_1	0.01437 04198
C_2	0.05928 01172
C_3	121.419 942

Exercise: What are the wavelengths of the resonances assumed in the model above? If you program this Sellmeier expansion, and evaluate it at wavelengths shown in the table above, do you really get agreement with the indices n given in the table above? (It is not quite trivial, even with 6 adjustable parameters, to fit 11 tabulated values to 5-digit precision – try some alternative fitting functions, if you wish.)

TeachSpin's experiment also comes with glass 'sample tubes' that permit the use of liquid (rather than solid) samples. This permits a wide variety of samples to be used – any transparent liquid that can be used to fill the tube can have its Verdet constant measured. Given the prediction of Section 1 that $V \propto n \cdot dn/d\lambda$, there's an appetite for $n(\lambda)$ -information for each liquid used. In principle, this can be measured via a hollow-prism experiment, but it's easier to rely on others' measurements of $n(\lambda)$ for a wide variety of samples. One source for $n(\lambda)$ information for many samples is refractiveindex.info, which gives empirical $n(\lambda)$ -information in both tabular and functional form.

3. Using the green-laser module

This Section assumes that you've used the Faraday-Rotation experiment in its original red-laser form, and that you now want to upgrade to the new green-laser module.

The upgrade is truly easy, as the new module has been built to be mechanically, optically, and electrically compatible with the red laser source. So before you do the change-over, get the red-laser source working, and notice how it gets its electric power. (Many users will see its red and black banana plugs going into back panel of a TeachSpin 'Power Audio Amplifier', which contains the 4-V special d.c. power supply built to supply power to these laser modules.)

[Other power supplies could be used, too. What you want is a voltage-controlled d.c. power supply that you can set to deliver a 4-V d.c. output, supplying a current up to 100 mA. The special requirement of the supply is that it *not* have any turn-on transient that takes the output voltage, even for a microsecond, much above the 4-V level. Similarly, you want an absence of transients above +4 V, or below 0 V, upon turning off the supply.]

Note that the green-laser module is shipped with its red and black power-input plugs shorted together. When you are ready to make the electrical changeover, turn off the power supply, pull out of the power supply the two leads for the red laser, un-short the leads for the green laser, plug its leads into the power supply *respecting polarity*, and then use the shorting wire to short the leads of the red laser, to protect it while it's in storage.

Now you can complete the mechanical changeover. Find the four white nylon thumbscrews in the laser mounting block on the Faraday Rotation apparatus, and back each screw outwards until it ceases to bear on its 'load'. Now withdraw the red laser module by pulling gently on its wire-bundle, drawing the inner block out of its larger mechanical enclosure. (That direction of pull is opposite to the direction of its output laser beam.)

Slide the green laser module into place by the reverse of the maneuver above. Now turn the four nylon thumbscrews inwards, until each of them bears on the inner laser block.

Once you have the green laser held in place this way, it is safe to turn on the power supply to check that you get a green output beam from it. Use a white paper card as a screen to confirm that the beam is present.

With the electrical and mechanical changeover complete, you can finish the optical change-over by aligning the beam. The tools to use are the four nylon thumbscrews; the goal is to get the green laser beam to pass cleanly into, and out of, the optical sample inside the solenoid in the apparatus. If you will be using electronic detection, you will also want the green laser beam, after passing through the rotatable Polaroid analyzer, to fall on the small dark square active area of the photodiode detector at the 'far end' of the optical bench of the apparatus.

4. Electronic detection of green light

The visual detection of the green laser beam on a screen or card presents no difficulties at all; in fact, your eye is even more sensitive to green light than to red. This Section deals only with the issue of using *electronic* detection of the green light, by sending it into the photodiode detector that's part of the Faraday-Rotation apparatus.

The photodiode works for green light just as it does for red: roughly speaking, each photon of light absorbed in its silicon-semiconductor surface liberates one electron-hole pair, and the electric field present in its p-n junction separates these charges to give a net external current that flows through the diode to some external circuit. You could show that this gives the photodiode a sensitivity for green light of about 0.5 mA/mW, a bit lower sensitivity than it has for red light.

In the TeachSpin set-up, that current flows through a resistor external to the photodiode, built into a little box. The value of the resistor can be set, via toggle switch, to 1, 3, or 10 k Ω , and you should pick the 1-k Ω setting for use with the full-intensity green beam. If you were to get 1 mW of green light onto the photodiode, you'd have about 0.5 mA of current flowing, and you'd get 0.5 V or 500 mV developed across the resistor, and sent down the cable to an external voltage readout.

In practice, the output voltage you get is very accurately linear in the incident optical power, *provided* the output voltage stays below about 300 mV. If you have a Faraday-rotation sample and an analyzing Polaroid in place in your optical 'beam line', you will probably stay within this 300-mV limit.

So if you have the green laser source installed, use the aiming adjusters at the source end of the apparatus to bring the beam-spot onto the small active area of the photodiode, and use a DMM to monitor the output voltage from the photodiode. You can adjust the beam position to maximize this output voltage; recall that you not only want the beam to fall squarely on the photodiode, but you also want it to enter, and to exit, the glass sample rod cleanly.

The use of the DMM provides a route to some of the electronic detection methods outlined in Section 5. But there is a surprise lurking under this DMM voltage, as you will see as soon as soon as you try an oscilloscope, instead of a DMM, as a detection tool.

A 'scope view of the signal from the photodiode reveals that surprise — the green laser beam is *not* a continuous steady beam (as it was in the red-laser case), but instead delivers its light in a series of *pulses*. Use the 'scope to find the average separation in time of these pulses; find the typical duration of the 'on' part of the pulse. Check to see if the peak of the pulse falls within the ≈ 300 -mV level, up to which the photodiode displays its best linearity.

What's going on here? It's all due to 'legacy effects'. Red diode lasers were, for a generation, the only visible diode lasers available. The semiconductors used to build them made it possible

to produce power-stabilized red diode-laser modules that operated using an external power supply of 3-4 V dc, and various markets developed that provided the voltage required by such modules. Green diode lasers depend on different semiconductors, and require a larger potential difference across their semiconductor junctions. To be able to market green-laser modules that would still work using the ‘legacy’ 3-4 V power supplies that red diode-laser users had come to expect, manufacturers used various tricks – and one of them has the result of pulsing the green laser.

You never noticed this by eyeball detection, because a fast enough succession of light pulses looks, to the eye, like a constant intensity. (This trick is *widely* used in general white-light LED illumination, as well as in LED screens, readouts, and displays.) But for oscilloscope detection, you’d like a signal that more closely resembles what you got from the red laser – one which is a constant.

An easy solution is to make your downstream electronics do what your eye does – namely, to have it *average* over some increment of time that contains many green-laser pulses, and to have it give an output voltage which is more nearly a constant reflecting the time-averaged level of green-light illumination.

That signal is easy to obtain using ‘low-pass filtering’. TeachSpin’s green-laser upgrade includes a little low-pass filter module, designed to be installed at the input of the ‘scope or DMM that you’re using to monitor the signal on the end of the photodiode cable. Send that cable into the filter’s INPUT end, and connect the OUTPUT end of the filter directly to the ‘scope’s input BNC connector. Now use the toggle switch to choose a suitable time-constant for the filter, and look for the result on the ‘scope.

The longer the time-constant you pick, the more averaging you will do, the smaller fluctuations you will see, and the slower the response to genuine changes will be. If you are using alternating-current excitation of the solenoid at frequency f , so that the green-light level is undergoing genuine modulation at that frequency, then you might want to pick the frequency f and the time constant τ of the filter such that $2\pi f\tau$ is reasonably small compared to 1; otherwise you would attenuate the a.c. signal you are trying to see.

This technique of low-pass filtering in particular, or filtering in general, is very widely used in signal transmission. You will not always have so clear a view of the unfiltered, and the filtered, signal, but you should learn the how, and the why, of this technique. More details about low-pass filtering are found in Appendix A1 of this Manual.

5. Some methods for observing Faraday rotation

This Section lists a series of methods that can be used in pursuit of Faraday-Rotation information. Any of them can be used with red, or with green, laser illumination. Some of them are optimal for demonstration and intuitive observations; others are more complicated, but also better suited for the detection of small Faraday rotations.

All of these methods assume that you have some linearly-polarized laser source in place, properly aligned, and that you have a solid or liquid-cell sample in the beam line. All of them also assume that the beam encounters a Polaroid analyzer after it exits the sample.

a. The 'protractor' method

This is the most direct and vivid way to show that there is such a thing as Faraday rotation. You will be rotating the Polaroid analyzer to find what is needed to 'extinguish' the transmitted beam, and then reading an angle (in degrees) by the orientation, relative to a reference line on the Polaroid, of an angular scale. Your diagnostic observable is the brightness, on a white card as screen, of the laser beam that emerges from the analyzer.

To perform these experiments, you'll need a d.c. power supply connected to the solenoid, and you will want first to set its output current to zero. You will form the 'control group' by finding an orientation of the analyzer that best 'extinguishes' the transmitted beam. Record this angular setting. (If you find a setting that does this, it should not surprise you that there is another setting, 180° away, that works just as well.)

Now dial the solenoid current to some chosen value, such as 1 A, 3 A, 6 A, or even 10 A. (The solenoid will tolerate long-term operation at 6 A, and short-term operation at 10 A. The limiting factor is the temperature to which the copper windings of the solenoid will rise.) When the current is on, the formerly extinguished beam will re-appear on the screen. Find the nearby orientation of the analyzer that extinguishes the beam again, and record its angular position.

Now dial the current to zero, (only then) interchange the solenoid's leads, and dial up the current to the same value formerly used. Find another 'extinction angle', this one to be found on the opposite side of the control-group angular position. Record this angle as well. Dial the current back down to zero.

You now have three angular settings, and can interpret their differences as Faraday-rotation angles. Recall that in each case you are inferring a direction of linear polarization of the light by using the angle to which a Polaroid has to be turned to extinguish that light.

If you are interested, you can even check Faraday's claim about the *sign* of the magnetic rotation: he observed that the direction of polarization-rotation is the same direction as the one that conventional electric current takes in wrapping around the solenoid.

The main limitation of this method is that Faraday-rotation angles are small, and the resolution of the angular scale you are using is not favorable. You can see 5°-increments directly, and can interpolate to perhaps 1° accuracy, on the angular scale. The most favorable experiments are those done with large currents, and with the larger Faraday rotations you might get using green light.

b. The 45°-methods

The methods below all make use of electronic detection, and all of them operate with the Polaroid analyzer oriented neither parallel to the light's polarization (where the maximal amount of it will be transmitted) nor perpendicular to the light's polarization (where the minimal amount will be transmitted – this is the extinction condition). Visually, this is very unspectacular method of operation, but using electronic detection it is very revealing.

The reason it works is due to the Law of Malus. We model the intensity emerging from an analyzer as

$$P(\theta) = P_{\max} \cos^2(\theta - \theta_{\max}) ,$$

where P stands for an optical power, and θ is a setting of the analyzer angle on the angular scale. (You might *read* that scale in degrees, but we will assume radian-measure of the angles above.) Here $P_{\max}$ means the maximum power that can emerge from the analyzer, when it is set to $\theta_{\max}$, the required angle for this (this $P_{\max}$ is, in fact, smaller than the total power that arrives at the analyzer). This model also assumes a 'perfect extinction', i.e. that the correct setting of θ can reduce the transmitted optical power all the way to zero.

Now your previous use of the analyzer is to find the angular reading θ_0 at which $P_{\max}$ is reduced to a P -value of zero; this occurred when θ_0 was $\pm 90^\circ$ away from $\theta_{\max}$, and it gave you evidence of the direction-of-polarization of the light reaching the analyzer. In practice, you found θ_0 by turning the analyzer so as to make the transmitted beam 'go away', and perhaps you 'dithered' the analyzer back-and-forth a bit to close in on the optimum setting of θ_0 . In the vicinity of θ_0 , the transmitted power behaved as

$$P(\theta_0 + \epsilon) = P(\theta_0) + \epsilon \cdot P'(\theta_0) + \frac{1}{2} \epsilon^2 \cdot P''(\theta_0) + \dots = 0 + \epsilon \cdot 0 + \epsilon^2 \cdot 1 \cdot P_{\max} + \dots$$

That is to say, there was *no* power coming through when you were right at θ_0 , and there were *no* first-order changes when you 'dithered' the analyzer around θ_0 ; there was only a weak *second-order-in- ϵ* increase in transmitted light on either side of the optimal extinction angle θ_0 . If you

had the analyzer set to better than half-a-degree to θ_0 , then you had $\varepsilon \approx 0.01$ radian or less, and second-order in ε implies *parts-in-10⁴* effects.

The new methods about to be introduced change these ε^2 effects into much larger ε^1 effects, and ones much more suited to electronic detection. Suppose we operate with the analyzer set, not 90° away from $\theta_{\max}$ (where extinction occurs), but 45° away from $\theta_{\max}$ instead. We get

$$P(\theta_{\max} \pm 45^\circ) = P_{\max} \cos^2(\pm 45^\circ) = P_{\max} \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2} P_{\max} \quad .$$

In fact, this result provides the easiest way to set to these angles: we read a voltage coming from the photodetector, one which would be $V_{\max}$ at optimal analyzer angle, and zero at the extinction angle; and we merely turn the analyzer to an angle at which the voltage drops to just *half* of $V_{\max}$.

Now we'll see what happens when the light reaches the analyzer not quite at 45° , but at $45^\circ + \varepsilon$. This can happen either when we turn the analyzer a bit (relative to light of a fixed polarization), or when Faraday rotation turns the polarization a bit (relative to a fixed analyzer). We get

$$P(\theta_{\max} + 45^\circ + \varepsilon) = P_{\max} \cos^2(45^\circ + \varepsilon) = P_{\max} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \varepsilon + \dots\right)^2 = P_{\max} \left(\frac{1}{2} + 1 \cdot \varepsilon + \dots\right)$$

In fact, relative to the power level of $P_{\max}/2$ (or relative to an output voltage level of $V_{\max}/2$) we now get changes; and better still, we get changes that are *first-order* in ε .

To be concrete about this, and to teach you that you can measure angles with a voltmeter(!), imagine a situation in which the largest photodiode signal you can get (for any orientation of the analyzer) is 250 mV; this is the $V_{\max}$ value. Then with the solenoid off, you rotate the analyzer to a position (which will be 45° to either side of this $\theta_{\max}$ position) at which the voltage coming from the photodiode is *half* this, i.e. it reaches 125 mV. Now suppose that the solenoid, when energized, gives to the light emerging from the sample a Faraday rotation of $\varepsilon = \pm 0.01$ rad. The model above predicts a photodiode-voltage *change* of

$$V(\theta_{\max} + 45^\circ + \varepsilon) = 125 \text{ mV} + 250 \text{ mV} \cdot \pm 0.01 = 125 \text{ mV} \pm 2.5 \text{ mV} \quad .$$

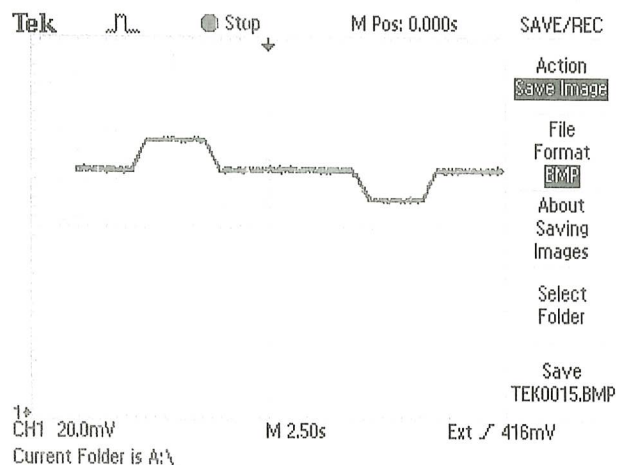
By contrast, if you had operated with the analyzer at the extinction condition instead, for these ± 0.01 -radian (or $\pm 0.57^\circ$) changes of polarization you'd have gotten just +0.025 mV for the voltage changes, an effect smaller by a factor of 100.

Note that if you measure $V_{\max}$ with a voltmeter, and you use it to measure changes relative to the 125-mV level, you can infer the sign, as well as the magnitude, of ε (in radians) via

$$\frac{V(\theta_{\max} + 45^\circ + \varepsilon) - V_{\max}/2}{V_{\max}} = \varepsilon \quad ,$$

so long as ε is small enough for this linear term to be the dominant term in the Taylor expansions we've been making.

Here's a very direct demonstration of this effect. Below we use an oscilloscope in the 'roll mode', at 2.5 s/div, so it slowly draws a horizontal line, at a vertical level set by a voltage coming into its input. We set up a Faraday-rotation apparatus and find that the $V_{\max}$ signal we can get is 250 mV. So we rotate the analyzer until this level drops to 125 mV. Now while the point on the 'scope-face is rolling along at the +125-mV level, we dial up the solenoid current to +5.5 A briefly, and then down to zero; then we interchange the leads to the coil, and dial up the power supply again, to give a -5.5-A current. All the while the 'scope is showing the V -signal we get from the photodiode. (The zero-voltage level is at the bottom of the screen shown here.)



What we see above is two excursions away from the +125-mV level, one of them an increase of about 15 mV, and the other a decrease of about 15 mV. We infer that the Faraday rotation caused by a 5.5-A coil current has a radian measure of

$$\varepsilon = \frac{15 \text{ mV of change}}{250 \text{ mV of } V_{\max}} = 0.060 \text{ (rad)} ;$$

Note that this is an angular change of 3.4° , which is less than one division on the angular scale of the 'protractor method' previously used. Our experimental resolution is no longer limited by the markings on the angular scale, but instead by the resolution of our 'scope or voltmeter.

c. Alternating-current methods

What works as above for brief excursions to positive or negative coil currents works just as well for sinusoidal alternating currents sent through the solenoid. The TeachSpin 'Power Audio Amplifier', for example, can turn an a.c. voltage sinusoid coming from some signal generator into an alternating current through the solenoid, of amplitude up to 1 A. This is less than can be attained using a d.c. power supply, but it still has advantages.

In particular, you can 'position' that current in frequency space, setting its frequency to (say) 37 Hz. Then the view above would be replaced by a steady 125-mV level, superimposed on which

would be a 37-Hz signal of amplitude about $(1/5.5) \times 15$ mV or 2.7 mV. It would be trivially easy to set the 'scope to 'a.c. coupling', which would block that steady 125 mV, but fully pass the 37-Hz sinusoid of 2.7 mV amplitude. With that change, one could set the 'scope to its maximum sensitivity so as to see the sinusoid optimally.

One way to optimize that view would be to trigger the 'scope on the same signal that's running the Power Audio Amplifier; that way, the Faraday-rotation signal would be a steady waveform on the 'scope, repeating itself trace after trace. You could speed up the time base to see its individual cycles; and best of all, you could now use the *averaging* mode of the 'scope to average away anything that was not a 37-Hz signal.

This is a powerful method of signal-averaging, very suitable for rejecting electronic noise, stray room light on the photodetector, and other forms of interference. It preserves the information on the shape of the waveform – in this case, sinusoidal.

If you trust that the waveform is sinusoidal, and if all you want to measure is the *amplitude* of the sinusoid that arises at the modulation frequency you're using to drive the coil, then the tool optimized for this purpose is called a 'lock-in amplifier'. It too needs, as a 'trigger' or synchronizing signal, a copy of the 37-Hz waveform from the signal generator. It too takes in the $V(t)$ signal from the photodetector, and it too is set for a.c. coupling to reject the steady $V_{\max}/2$ part of the signal. But a lock-in amplifier can easily be set to detect not just millivolt-level signals, but even signals of a microvolt, or even 100 *nanovolt* (10^{-7} V), amplitude. Hence Faraday rotation signals of vastly smaller size than discussed heretofore can easily be detected.

More details of the lock-in detection method may be found in a tutorial article at the TeachSpin website. On the page <http://www.teachspin.com/signal-processor-lock-in-amplifier.html>, find the Additional Resources box, and go to 'A case study in small-signal measurement'. This reference will show you how a lock-in amplifier can be used, quantitatively, to measure Faraday-rotation angles much smaller than a milliradian of angle. Along the way, you'll also learn some transferable skills of how to use a lock-in.

Appendix A1. Low-pass filtering – a case study

If you have had a course in electronics, or perhaps just in differential equations, then you know that a simple circuit involving a resistor and a capacitor acts like a ‘low-pass filter’. It’s worth understanding this simple circuit from multiple viewpoints, since it will give you a vocabulary that is applicable to a host of similar problems in signal processing.

We can draw the circuit as involving a voltage source V_{in} and giving a voltage output V_{out} . In practice, we have an input $V_{in}(t)$, perhaps a signal whose average level we want to measure, but bearing fluctuations that we want to ‘average out’; and we hope to get an output $V_{out}(t)$ with properties more to our liking.

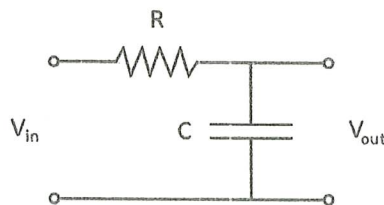


Fig. A1: The simplest RC-filter circuit.

You should be able to derive the differential equation describing $V_{out}(t)$ in terms of $V_{in}(t)$; one tool for deriving it is to introduce the charge $q(t)$ present on the capacitor. If we neglect any current drawn at the output of the circuit, the result can be written as

$$RC \frac{dV_{out}}{dt} + V_{out}(t) = V_{in}(t) \quad .$$

Your intuition ought to show you that if $V_{in}(t)$ is a constant, then $V_{out}(t)$ eventually reaches this same constant value. Furthermore, if $V_{in}(t)$ changes from an old constant to a new constant (i.e. if it undergoes a ‘step change’), then $V_{out}(t)$ changes from its previous old value, and approaches its new value – but not ‘all at once’. The response to a step-change at the input is an ‘exponential decay’ toward a new level at the output, with the output voltage asymptotically approaching the eventual new level. The ‘distance to go’ to the eventual asymptotic value drops by a factor of e for every elapsed time interval of duration τ , where τ is called the time-constant and is given by $R \cdot C$.

But you should have *more* ways of understanding this circuit than only via the step-change response just discussed. One other viewpoint you should have is the ‘frequency-domain response’, which answers the question – what is the response of this circuit to an input which is a sinusoidal function of time? You should be able to show that the output is also sinusoidal, and of the same frequency as the input, but attenuated (as well as phase-shifted). For frequency of excitation f , you should be able to show that the amplitude A_{out} of the output sinusoid is related to the amplitude A_{in} of the input sinusoid by

$$\frac{A_{out}}{A_{in}} = \frac{1}{\sqrt{1+(2\pi f \tau)^2}} \quad .$$

The two relevant limiting cases here are low frequency [$f \ll 1/(2\pi \tau)$] where $A_{out} \approx A_{in}$, and high frequency [$f \gg 1/(2\pi \tau)$] where $A_{out} \ll A_{in}$. In fact, high-frequency components of the input are attenuated in magnitude by a factor near $(2\pi f \tau)$, and this is a large attenuation for large enough f -values. So we speak of the filter ‘blocking’ the high frequencies, and ‘passing’ the low frequencies, even though in reality it’s not quite such an all-or-nothing operation. ‘Low-pass filtering’ means that low (enough) frequencies are passed-through with little change; and (implicitly) means that high enough frequencies are not passed through, but instead are blocked.

This viewpoint is great if you can understand $V_{in}(t)$ in ‘Fourier terms’. In the context of our green-laser signal, the signal we want is the ‘d.c. average’ or the zero-frequency Fourier component, while the signal we’d like to filter *out* lies at about 40 kHz and all its harmonics – that’s what it takes, in Fourier terms, to represent the periodic pulses the laser produces. So we want a ‘corner frequency’ of the filter to be somewhere well above $f=0$, but well below $f=40$ kHz.

There’s one more kind of understanding you can attain about this filter, and it comes from a time-domain view of $V_{out}(t)$. Using the ‘integrating factor’ or Green’s-function technique, you can easily solve the first-order, linear, inhomogeneous differential equation above to get the (exact) solution

$$V_{out}(t) = \frac{1}{\tau} \int_{-\infty}^t e^{-(t-t')/\tau} \cdot V_{in}(t') dt' \quad .$$

This says the present value of V_{out} , namely $V_{out}(t)$, is given by an integral over *past* values of $V_{in}(t)$: the integration variable t' in the integral above runs from $t' = -\infty$ only up to the present time t , *not* into the future. The various values of $V_{in}(t')$ in the past get ‘weighted’ differently in their contribution to the integral. The most recent values of t' , the ones with t' just less than t , have a relative weight near 1. Values of $V_{in}(t')$ that are ‘one-time-constant ago’ have $t' = t - \tau$, and they have a relative weight of e^{-1} . Values of $V_{in}(t')$ in the more distant past are exponentially de-weighted in the integral. The factor of $1/\tau$ in front ensures the sum-of-weights is just one, so any constant V_{in} gives, as output, the *same* value of V_{out} .

The net effect is that $V_{out}(t)$ is a kind of ‘running average’ of $V_{in}(t)$ -values from the recent past. The ‘memory depth’ of that weighting is given by the time-constant τ . Any transient feature in $V_{in}(\text{past } t)$ affects $V_{out}(\text{present } t)$, but the size of that transient’s effect decays exponentially with the distance-into-the-past of the transient’s occurrence.

Now you have *three* viewpoints on this filter, and three vocabularies to use in understanding any filtering operation. You understand the step-response, the frequency-domain response, and the time-domain response. These are vocabulary terms and mental tools of applicability far beyond Faraday rotation!

Now, about the actual design of the filter. Its only components are a (fixed) resistor and a single capacitor (from a choice of three). If you were the designer, you'd have to pick an R -value and a C -value. Below is a discussion of how *two* experimental considerations allow you to 'solve' for these two unknowns.

We have a laser which operates in a pulsed mode, producing a train of optical 'bursts', each of about ten microseconds' duration, and repeating every 25 μ s or so, so that pulses occur at a frequency of about 40 kHz. But what we see by eyeball viewing, and what we care about in electronic detection, is the laser's time-averaged output of optical power. Specifically, we'd like to form, electronically, the time-average of the output signal from a photodetector that views this laser light.

So we'll time-average over a time-scale τ which is *long* compared to the 25- μ s pulse repetition time of the laser, yet *short* enough to give us quick response to changes in the average intensity. To achieve this we can use a simple, passive RC-filter, made as in Fig. A1 out of one resistor and one capacitor. In the time domain, that gives as output a running-average over a time-scale of about $\tau = RC$ into the recent past. In the frequency domain, that gives nearly full response at low frequencies, half-power response at corner frequency $f_c = 1/(2\pi\tau)$, and a response dropping as f^{-1} above this 'corner frequency'.

We might choose $f_c = 400$ Hz, so that the 40-kHz fluctuations would be heavily (100-fold) suppressed. But honest-to-goodness variation of the average light signal, even at 37 or 137 Hz, would be only slightly attenuated. That choice gives $2\pi \cdot f_c \approx 2.5 \times 10^3$ /s, and it gives for τ the result $1/(2\pi \cdot f_c) \approx 0.4 \times 10^{-3}$ s = 0.4 ms or 400 μ s. Hence the R and the C we pick would be constrained to have a product $R \times C = 0.4$ ms.

But that's only one equation in two unknowns; what determines the R - and C -values individually? The separate values we need come from consideration of *impedances*. We have a light source (the photodiode and its load resistor) with output impedance in the 1 - 10 k Ω range, and we imagine loading the output of our filter with a 'scope or DMM, with input impedances 1 M Ω or 10 M Ω . To build a filter which scarcely 'loads down' the source, and which in turn is scarcely 'loaded down' by the destination, we'd like the filter's input impedance to have an intermediate value, so we might choose a sort of geometric mean at $R = 100$ k Ω .

And that choice would fix C to be $(R \cdot C)/R = (0.4 \text{ ms})/0.1 \text{ M}\Omega = 4$ nF. Our parts-bin might offer us a 3900 pF or 3.9 nF or 0.0039 μ F capacitor, and we'd be all set.

Notice the use of various desiderata, and engineering constraints, have separately determined both the R - and C -values we need. Note that corner-frequency alone did *not* fix the design for us; we needed also to consider impedances.

The TeachSpin Low-Pass Filter module that comes along with the green-laser accessory uses the choice $R = 100 \text{ k}\Omega$ mentioned above. The three positions of its toggle switch allow the selection of a C -value of either 0.22 nF , 4.9 nF , or $0.10 \text{ }\mu\text{F}$ to achieve time constants near 0.022 ms , 0.49 ms , or 10 ms .

Exercise: To what 'corner frequencies' do these correspond?

Appendix A2. Laser wavelengths

If your past acquaintance with lasers has been confined to gas lasers (such as helium-neon, or argon-ion, technologies), then you might think that every laser operates on the wavelength of some atomic spectral line. For example, the most familiar (red) He-Ne laser operates on a certain spectral transition in neon, at wavelength (in standard air) of 632.8 nm.

But diode lasers are different – the gain medium is *not* nearly-free atoms in the gas phase. Instead, the quantum transitions responsible for the laser action occur in solid-state semiconductors, and the actual laser wavelength depends (mostly) on the band gap of the semiconductor used. That, in turn, depends on what semiconductor family is used; next, on what alloying is used; also, on the temperature of the semiconductor; and finally, on the steady current that is passing through the diode junction of the semiconductor. The result is that the actual wavelength of a diode laser's output depends much more on technology than on atomic physics.

Manufacturing practice settles on some 'standard' wavelengths. The first commercialized diode lasers made CD and CD-ROM devices possible, and (for technical convenience) they were AlGaAs devices operating near 780 nm. But even in so mature a technology, manufacturers will still sell devices only with the claim of operation (under standard conditions) somewhere in the range 785 ± 5 nm.

The red diode laser that's part of Faraday Rotation belongs to another technology, and the manufacturers typically call one of these families 650-nm devices; in practice, they are specified to operate somewhere in the range 650 ± 10 nm.

So-called 'direct green lasers' are a more recent technology still; the one included in TeachSpin's 'green laser upgrade' is manufacturer-specified to operate in the range 520 ± 10 nm.

Now a particular single laser, operating at a fixed temperature and held at a fixed laser-output power, is *not* a broadband source as the numbers above might suggest. Unlike a green LED, which might very well produce light over an entire band of wavelengths, from 500 to 540 nm, a green diode laser in operation will operate at one particular wavelength (or, at worst, at a few closely-spaced wavelengths). That wavelength is going to fall somewhere in the range 510 to 530 nm, and the actual value could be determined (if desired) to three- or four-digit accuracy.

The best method for finding the actual operating wavelength of a given laser would be to calibrate an optical spectrometer (typically by the use of gas-discharge tubes, i.e. again atomic-spectroscopy standards), and then to get laser light into that spectrometer.

Technical Tip: So bright is laser light (compared to standard non-laser sources) that you should use a method cleverer than merely shining the laser beam into the spectrometer. One useful method is to put a sheet of ordinary white paper, or waxed paper, right onto the entrance of the spectrometer, and then to shine the laser onto that piece of paper. Not only does the paper attenuate the laser down to power levels safe for the spectrometer and its detector, this method also scatters the laser beam, forming a more omni-directional source, some of whose light is sure to get into the spectrometer propagating in the correct direction.

Note that the wavelength you find by this sort of method is *not* some invariant physical property of atomic energy levels. In practice, a diode-laser module's output wavelength will change with the temperature of the diode laser used, at a rate of about 0.3 nm/K – but the rate is not uniform, and the change might not even be monotonic. Hence it is not likely to be worthwhile trying to measure the 'exact' wavelength of your laser to better than 1-nm precision.

Note also that this method does not measure the optical power output of your diode-laser module. Circuitry in the module does keep this power level stable despite changes in the device's temperature. The TeachSpin red and green diode-laser modules have been chosen to have output powers just under 1 mW, to keep them in the least-restrictive regulatory class.

Appendix A3. The effective sample length

To make the detection of Faraday Rotation possible in more samples than just a glass rod, TeachSpin supplies a glass cell, which can be filled with any transparent liquid, to permit the measurement of its Verdet constant. The optical and/or electronic techniques you'd use are identical for a liquid cell as for a solid sample; the only complication is that the cell is *too long* to be treated as if it placed all of the sample in a constant-field region near the center of the solenoid. So this Appendix teaches you how to cope with the spatial variation of the magnetic field B along the length of an axis along the light beam. Formerly we took B as constant, and having value $k \cdot i$, for the full length L of the (glass) sample. Now we'll use coordinate z along the axis of the solenoid, and assume the solenoid gives a field variation with z , and we'll write that 'field profile' by assuming

$$B \rightarrow B(z) \equiv B_{center} \cdot \frac{B(z)}{B_{center}} = k \cdot i \cdot f(z) .$$

So by definition of k , the central field B_{center} has magnitude $k \cdot i$. [Because of the finite length of the solenoid, this will be smaller than the result derived from the infinite-solenoid approximation.] The function $f(z)$ is defined so as to have value 1 at the center of the solenoid, and values just below 1 in its central section; but the f -values drop to about $\frac{1}{2}$ at the ends of the solenoid's windings, and drop further still beyond those ends. Note that $k \cdot i$'s value, and $f(z)$'s shape, could be computed analytically from the theory of solenoids, or they could be obtained from 'surveying a map' of $B(z)$ -values, obtained (for example) with a Hall-effect probe. But what would we *do* with $f(z)$ profile information, to enable the computation of a Verdet constant for a long sample? Here's the use of it:

Formerly we assumed Faraday rotation $\theta = V B L$ for a sample of length L , and clearly that result specializes, for length dz of sample, to rotation $d\theta = V B dz = V B(z) dz$. But using our field profile information, we can write that as $d\theta = V \cdot k \cdot i \cdot f(z) \cdot dz$. Now we integrate over the entire region of the sample $\{S\}$ to get net Faraday rotation

$$\theta = \int_S d\theta = \int_S V \cdot k \cdot i \cdot f(z) dz = V \cdot k \cdot i \int_S f(z) dz .$$

Formerly, we assumed $f(z)$ was constant, and of value 1, over the length L ($=0.100$ m) of the glass sample, so this integration gave

$$\theta = V \cdot k \cdot i \int_S 1 dz = V \cdot k \cdot i \cdot L .$$

Now instead we get

$$\theta = V \cdot k \cdot i \int_S f(z) dz \equiv V \cdot k \cdot i \cdot L_{eff} .$$

so clearly all of our old results for data reduction hold true, provided we replace references to L by L_{eff} , where L_{eff} is the ‘effective length’ of the sample, given by

$$L_{\text{eff}} \equiv \int_S f(z) dz .$$

Notice the limits of integration are over the length of the sample S (*not* over the length of the solenoid). Notice that $f(z)$ is dimensionless, with value 1 at the solenoid’s center, so L_{eff} has dimensions of length arising from the integration of dz .

[If a sample were to be *really* long, extending far beyond both ends of the solenoid, the integral we need is related to one from Ampere’s Law, since for an N -turn solenoid we’d have

$$\int_{-\infty}^{\infty} B(z) dz = \oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 N i .$$

The Amperean path C has to be ‘closed at infinity’ for this to work, but a ‘semicircle at infinity’ gives zero contribution, and the whole of a closed-path integral is thus given by an integration on an infinite line, our z -axis. In our model for field profile, this gives

$$\int_{-\infty}^{\infty} k i f(z) dz = \mu_0 N i ,$$

from which we could extract

$$L_{\text{eff}} = \int_S f(z) dz = \frac{\mu_0 N i}{k i} = \frac{\mu_0 N}{k}$$

for a really long sample. In the case of the TeachSpin solenoid, this would give

$$L_{\text{eff}} = \frac{\mu_0 N}{k} = \frac{(4\pi \times 10^{-7} \text{ T}\cdot\text{m/A})(1400)}{0.0115 \text{ T/A}} = 0.1530 \text{ m} .$$

In practice, the liquid-sample tube does not extend so far as $\pm\infty$, and thus its value for L_{eff} will be shorter than this, and certainly shorter than the tube’s own physical length. (Notice that the number for L_{eff} in the very-long-sample limit differs from the physical length of the solenoid, 6” or 0.1524 m. Note also that it’s the $k \cdot L_{\text{eff}}$ product that you would need to extract a Verdet constant from your data, and in the very-long-sample limit, that product reduces to $\mu_0 N$, which requires no metrology except turn-counting!)]

